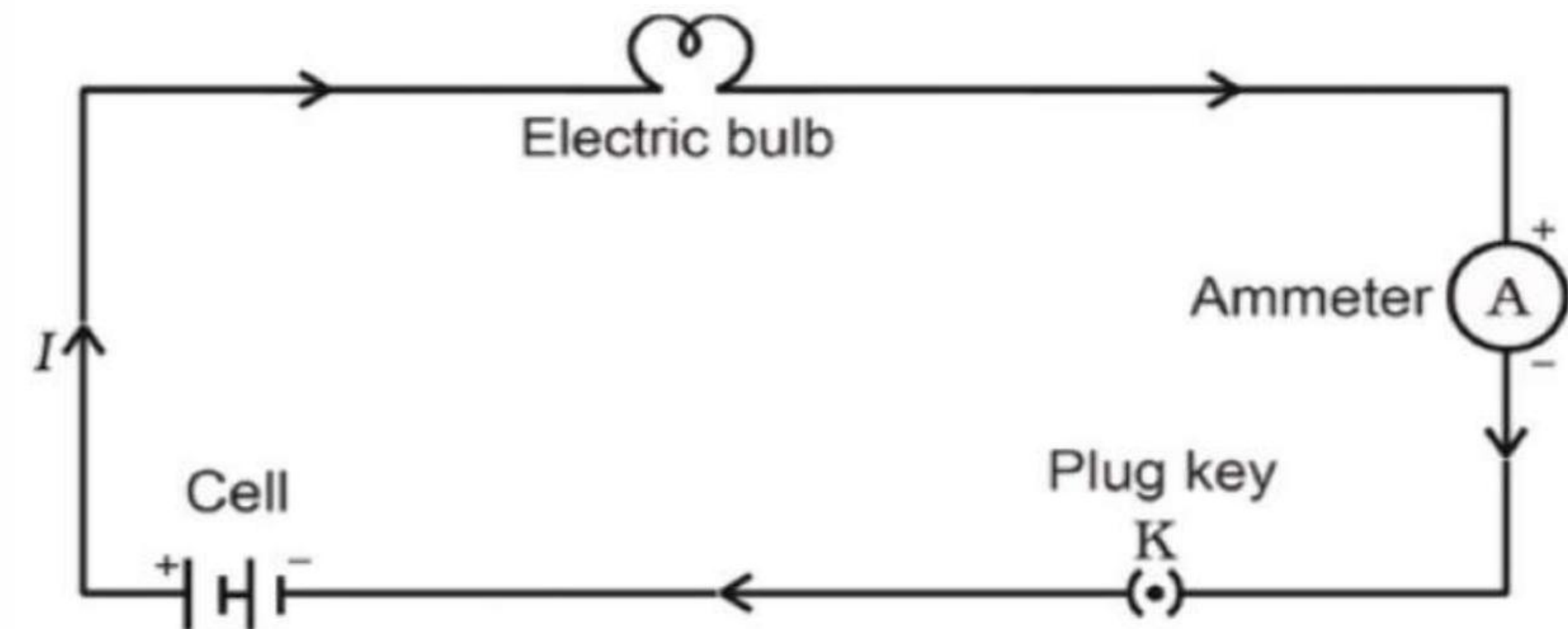
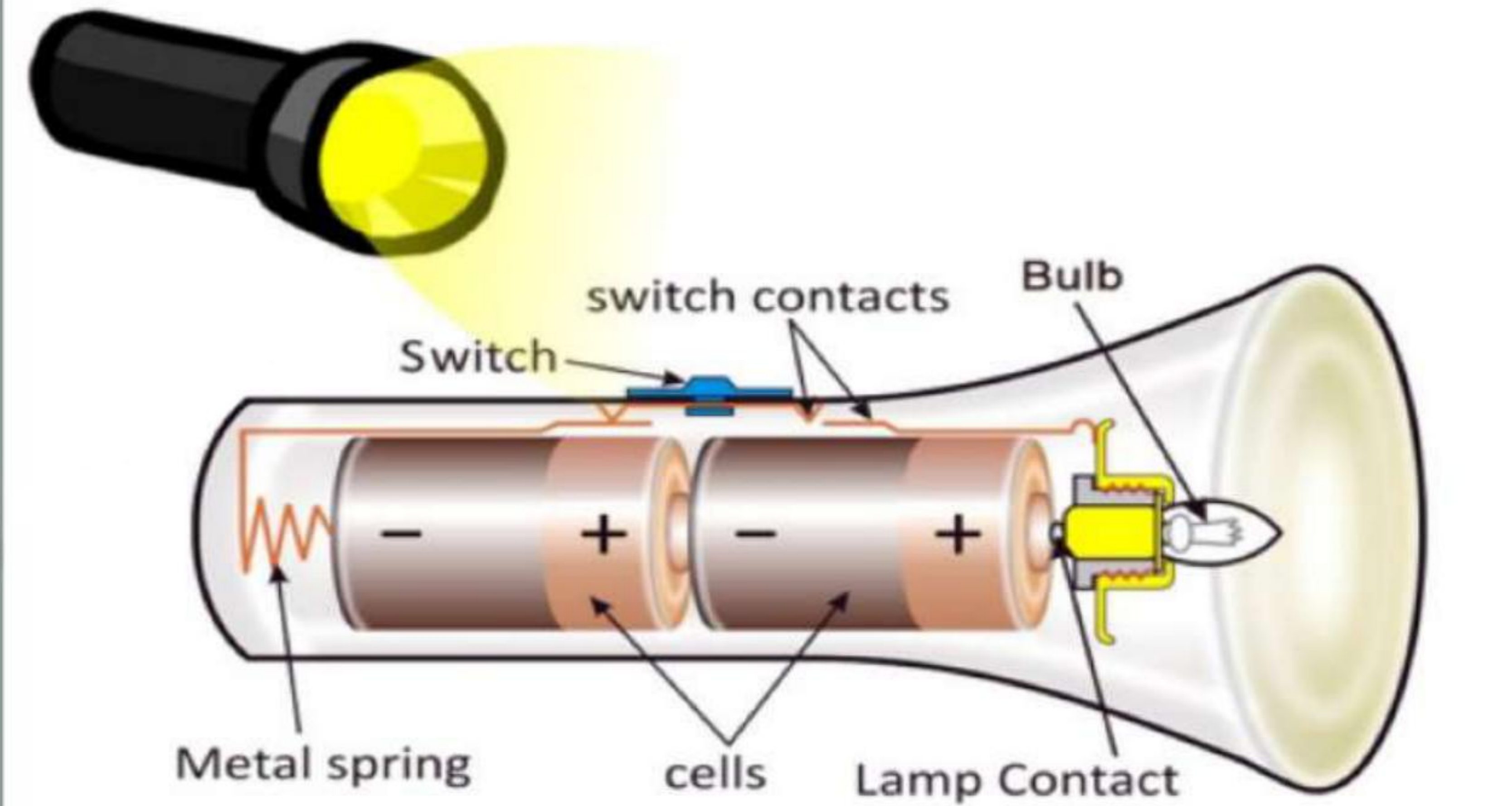




ELECTRICITY

ELECTRIC CURRENT AND CIRCUIT

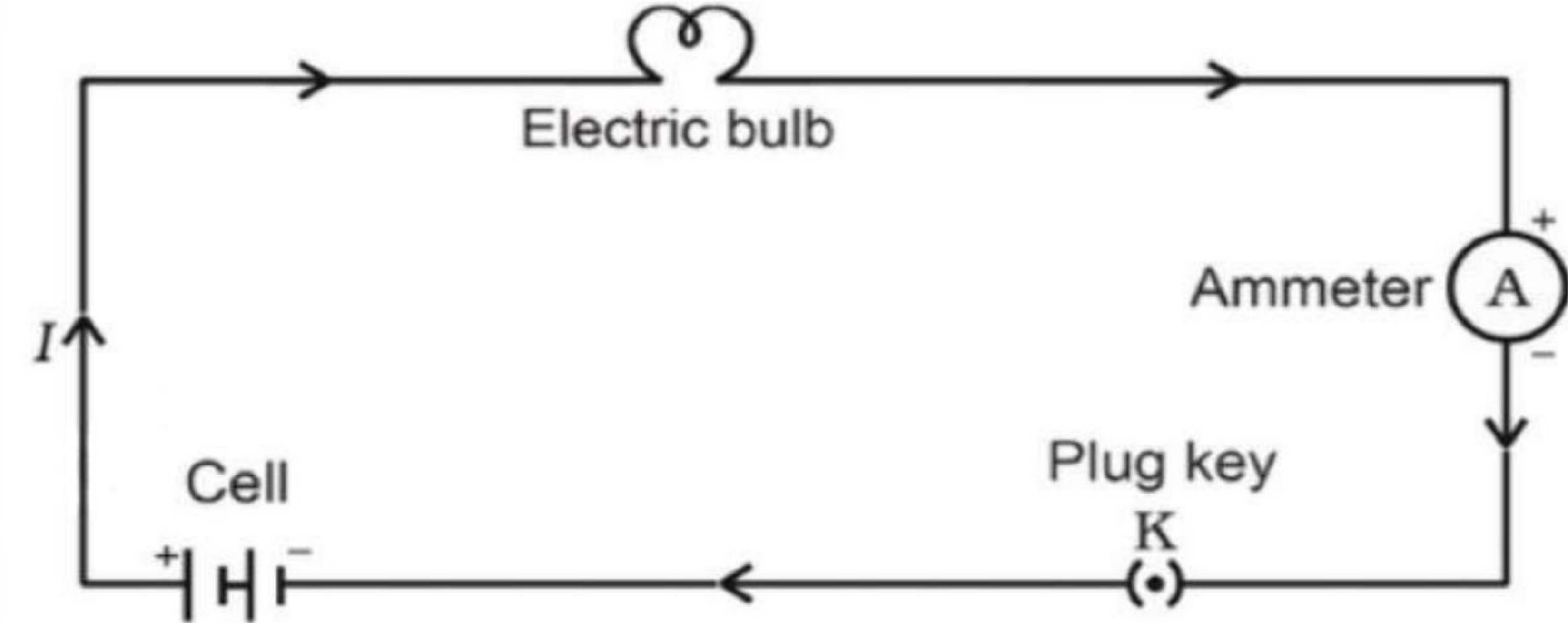
- Flow of electric charges through a conductor (e.g., metallic wire) is called an **electric current**.
- In a torch, the cells (or a battery, when placed in proper order) provide electric current through the bulb to glow.
- A continuous and closed path of an electric current is called **electric circuit**. If the circuit is broken (or the switch is turned off), the current stops.
- Electric current is measured as the amount of charge flowing through a particular area in unit time. i.e., it is the **rate of flow of electric charges**.



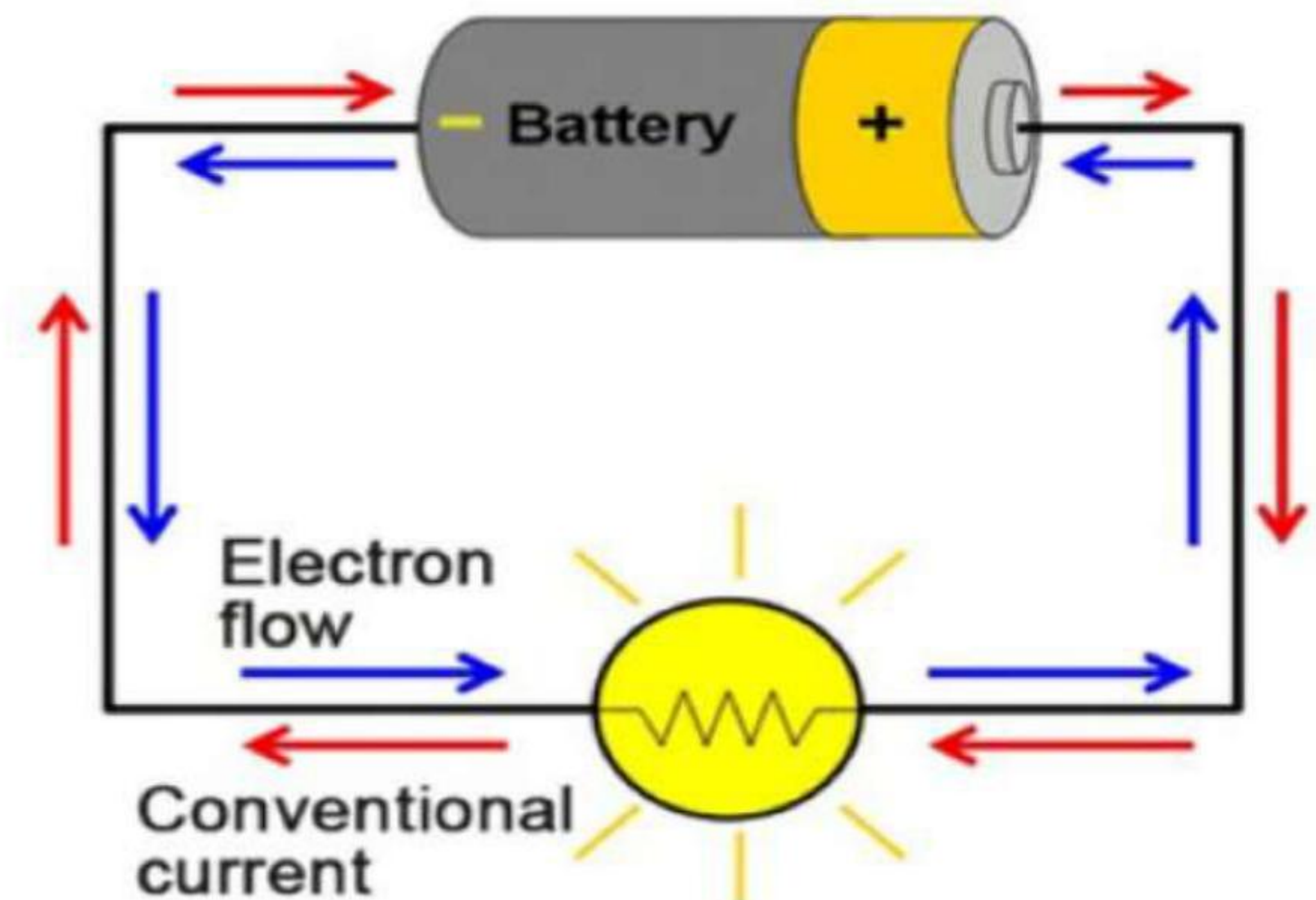
A schematic diagram of an electric circuit

ELECTRIC CURRENT AND CIRCUIT

- In a circuit of metallic wire, **electrons** constitute the flow of charges. Electrons were unknown during the discovery of electricity. So, electric current was considered as the flow of positive charges and their direction of flow was taken as the direction of electric current.
- Conventionally, in an electric circuit, the direction of **electric current is taken as opposite to the direction of the flow of electrons** (negative charges). i.e., electric current flows from positive terminal of the cell to negative terminal.



A schematic diagram of an electric circuit

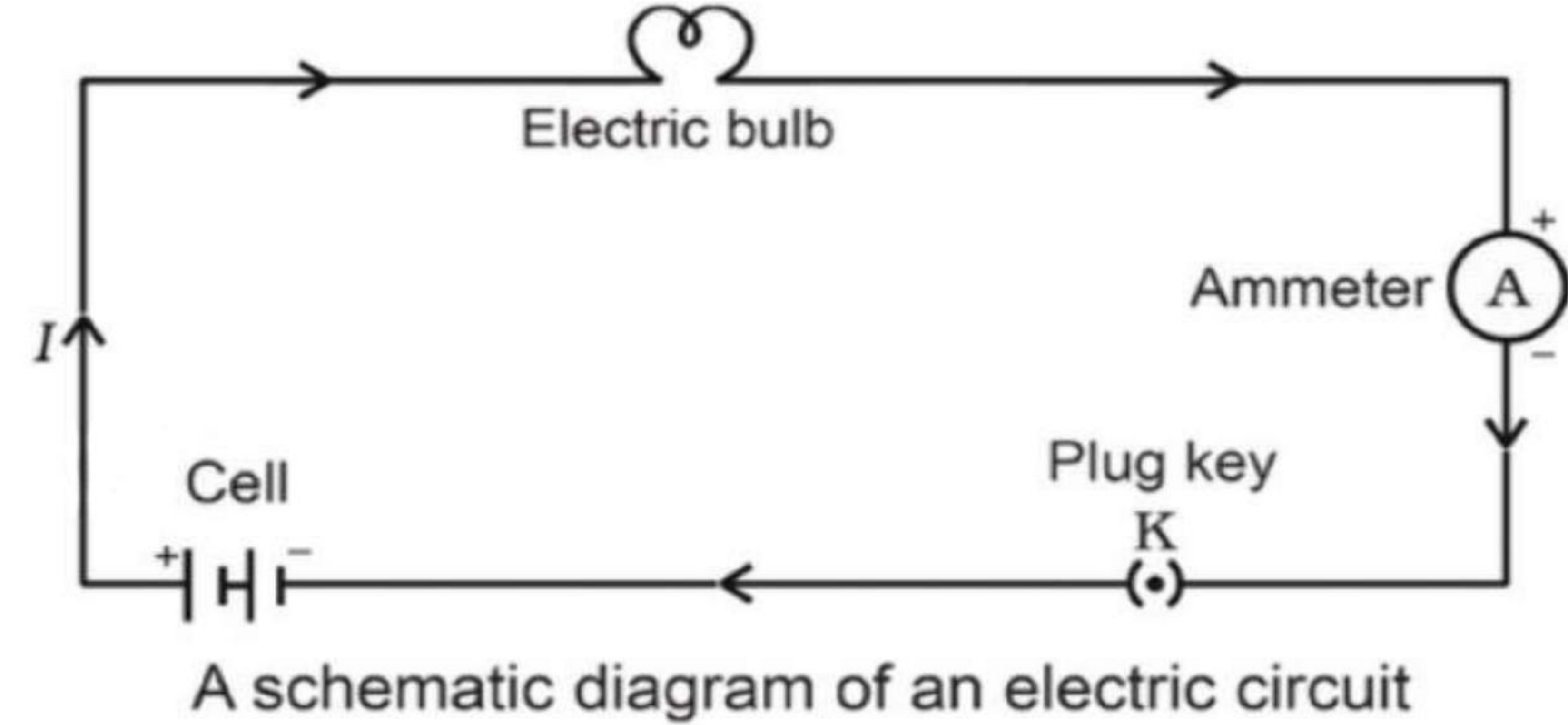


ELECTRIC CURRENT AND CIRCUIT

- Consider a **net charge Q** flows across a cross-section of a conductor in **time t** . Then, **current I** through the cross-section is

$$I = \frac{Q}{t} \quad \text{or} \quad Q = It$$

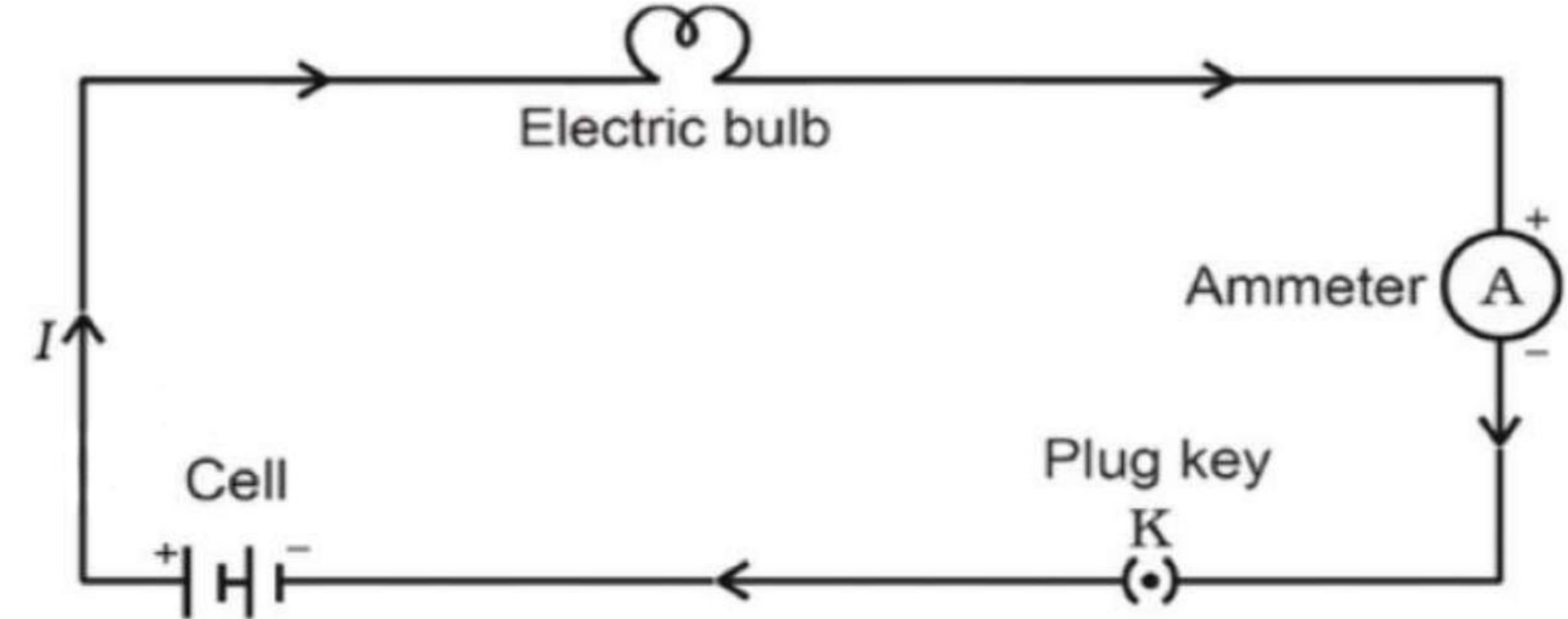
- SI unit** of electric charge is **coulomb (C)**. It is equivalent to the charge contained in 6×10^{18} **electrons**. i.e., an electron has a negative charge of 1.6×10^{-19} C.
- SI unit of electric current is **ampere (A)**, named after **Andre-Marie Ampere** (France, 1775–1836).



**Andre-Marie
Ampere**

ELECTRIC CURRENT AND CIRCUIT

- **One ampere** represents the flow of one coulomb of charge per second, i.e., $1\text{ A} = 1\text{ C}/1\text{ s}$.
- Small quantities of current are expressed in **milliampere** ($1\text{ mA} = 10^{-3}\text{ A}$) or in **microampere** ($1\text{ }\mu\text{A} = 10^{-6}\text{ A}$).
- **Ammeter:** An instrument to measure electric current in a circuit. It is always connected in series in a circuit.



A schematic diagram of an electric circuit



ELECTRIC CURRENT AND CIRCUIT

Problem

A current of 0.5 A is drawn by a filament of an electric bulb for 10 minutes. Find the amount of electric charge that flows through the circuit.

Solution

$$I = 0.5 \text{ A}; t = 10 \text{ min} = 600 \text{ s.}$$

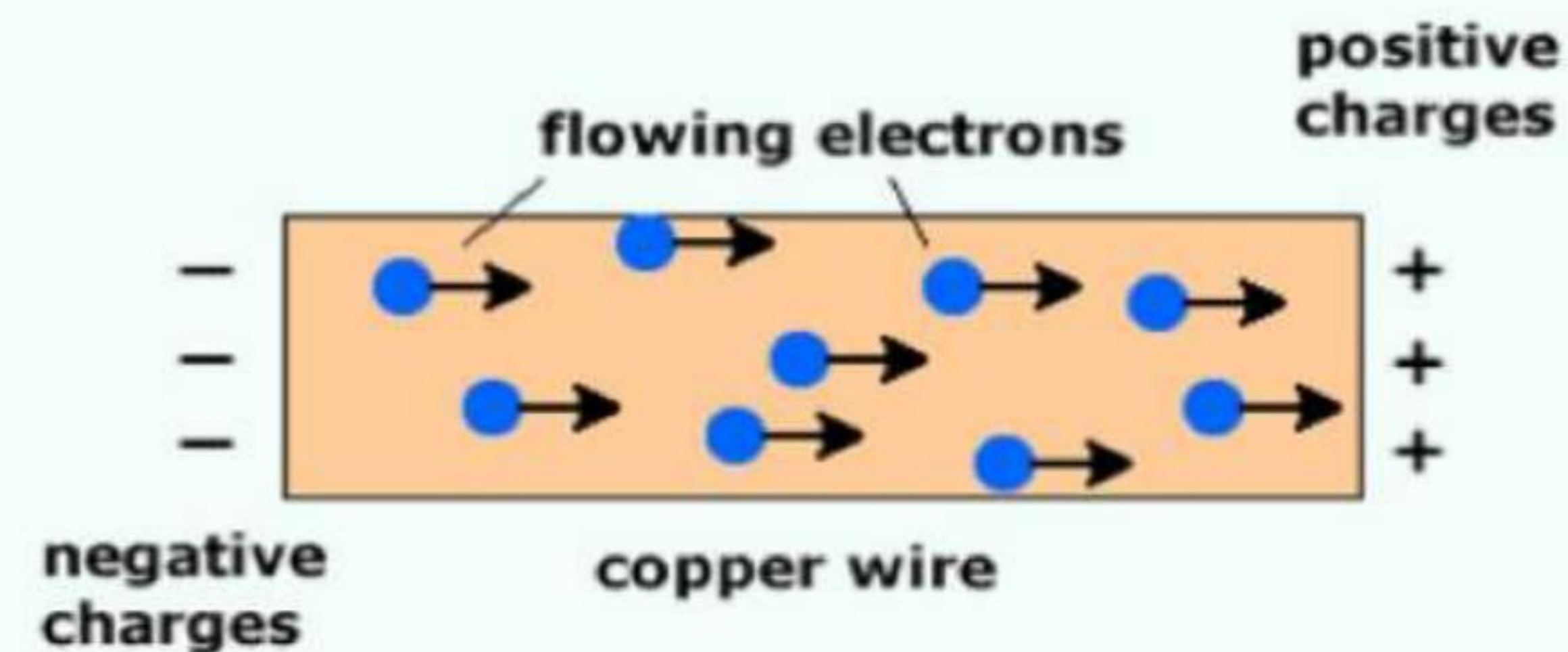
$$Q = It$$

$$= 0.5 \text{ A} \times 600 \text{ s}$$

$$= \underline{\underline{300 \text{ C}}}$$

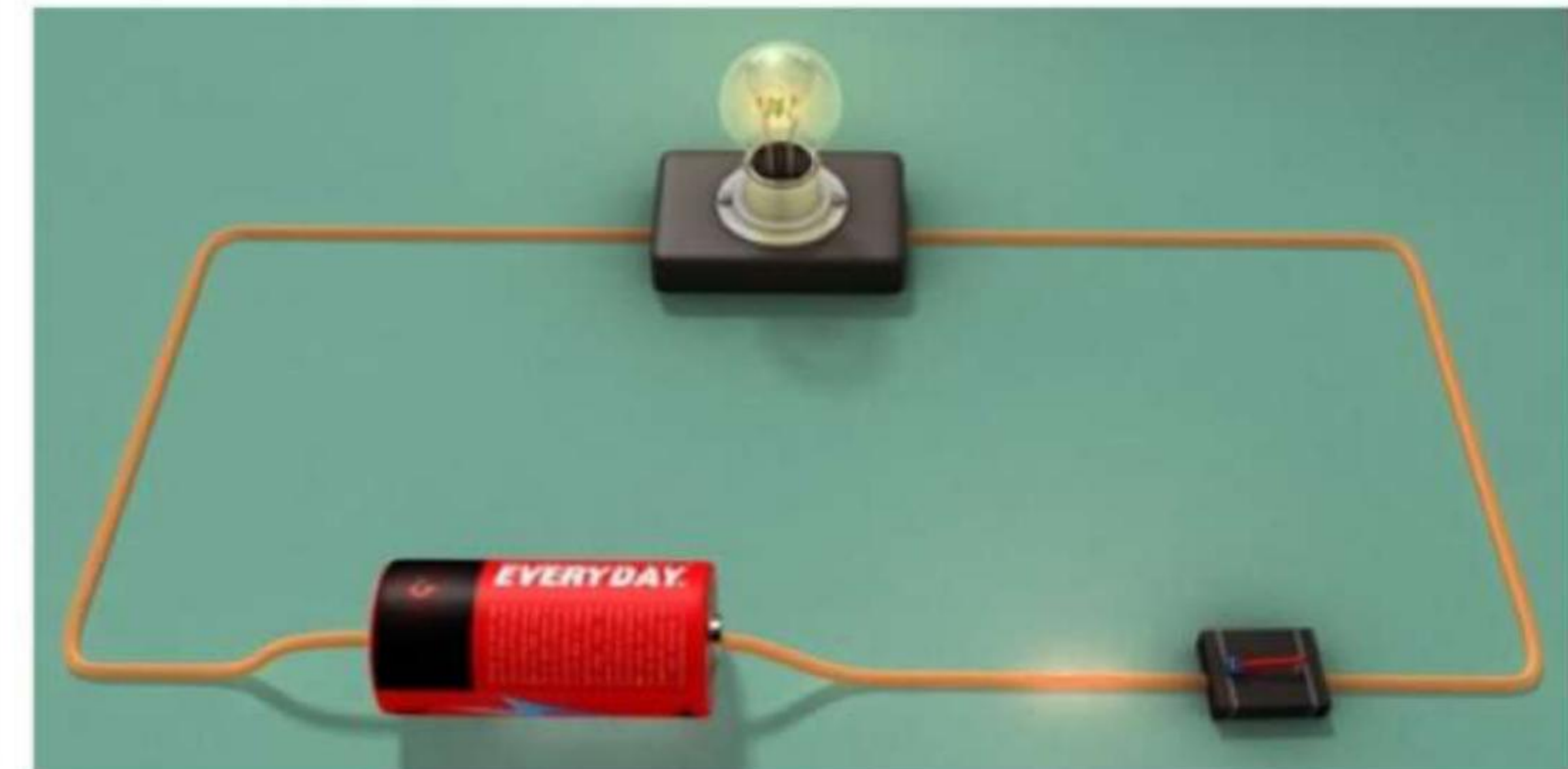
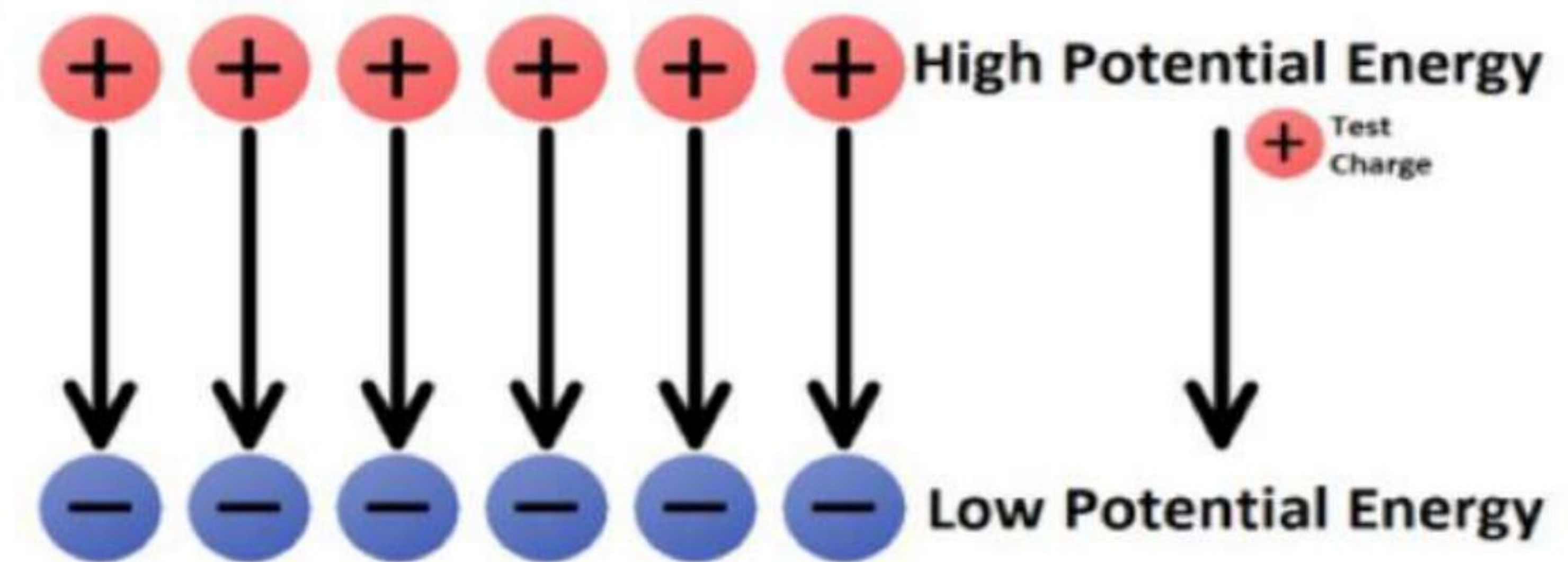
'Flow' of charges inside a wire

- Inside a solid conductor, atoms are packed together. But electrons can easily travel through it, almost as if they are in a vacuum.
- When a current flows through a conductor, the electrons move with an average 'drift speed' (1 mm s^{-1} for a typical copper wire). However, an electric bulb lights up as soon as the switch is turned on. It is not due to the movement of an electron from one terminal to the other terminal through the bulb.



ELECTRIC POTENTIAL AND POTENTIAL DIFFERENCE

- Electric charges do not flow in a metallic conductor (e.g. copper wire) by themselves. Here, **gravity has no role**.
- Electrons move only if there is a **difference of electric pressure** in conductor. It is called **potential difference**.
- The chemical action within a **cell/battery** can generate the potential difference across the terminals of the cell, even when no current is drawn from it.
- When the cell is connected to a conducting circuit, the potential difference sets the charges in motion and produces an electric current. To maintain this current, the cell has to expend chemical energy stored in it.



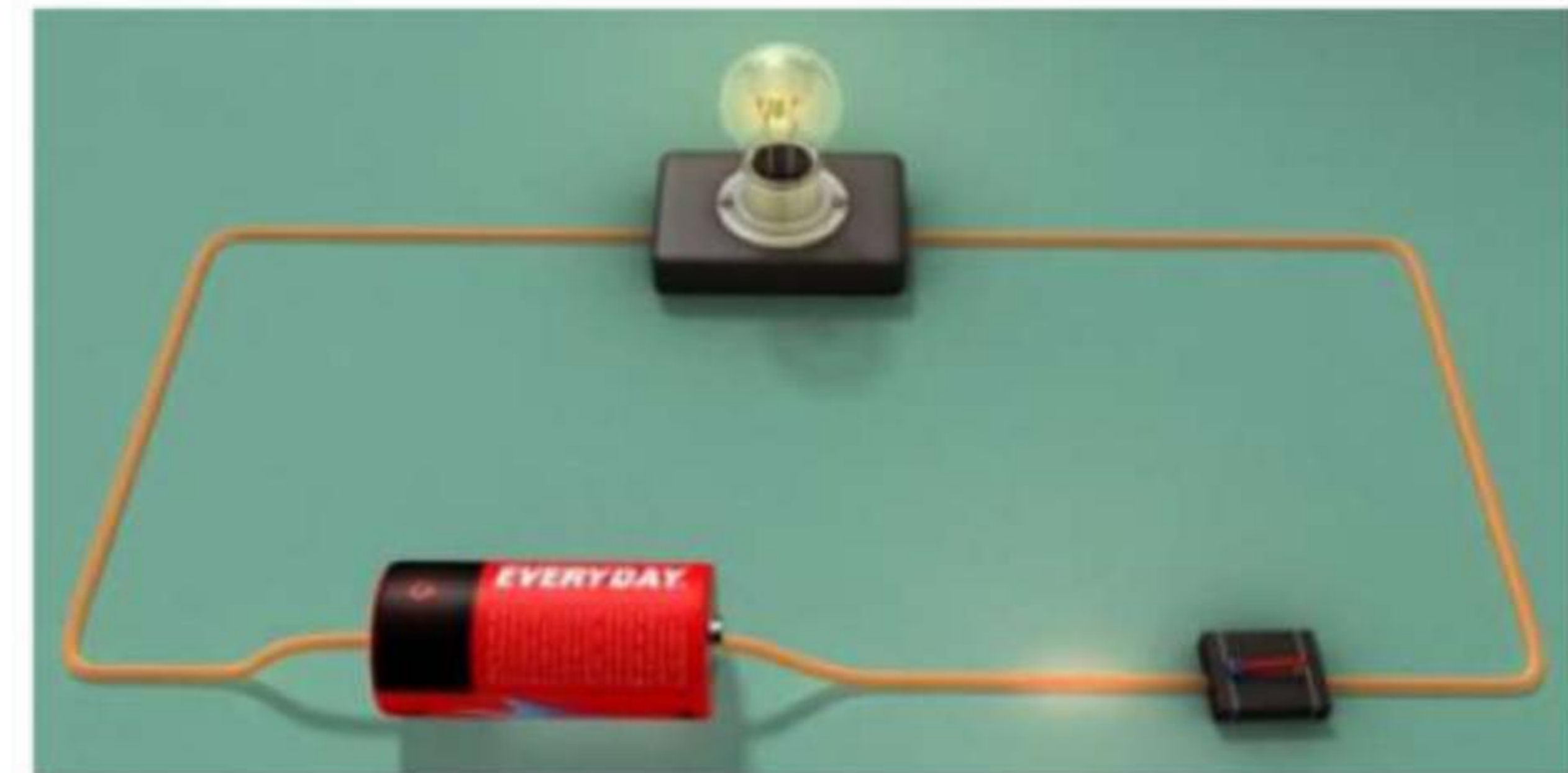
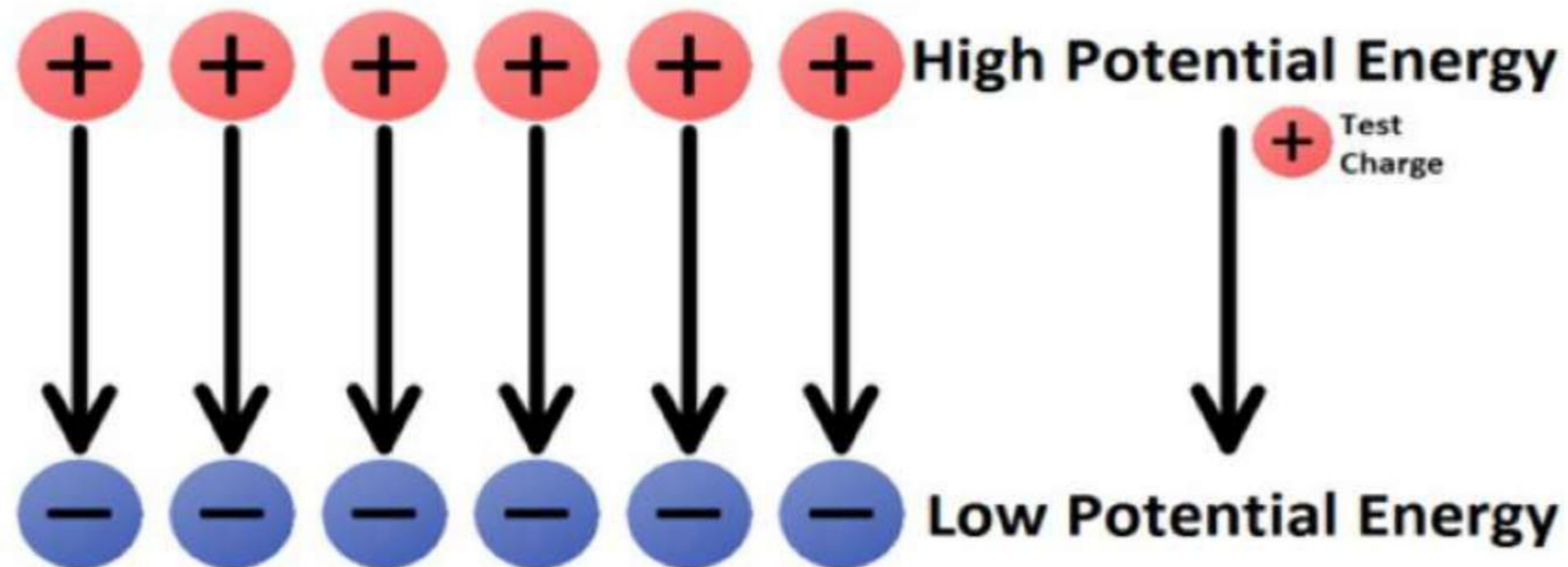
ELECTRIC POTENTIAL AND POTENTIAL DIFFERENCE

- The **electric potential difference** between two points in an electric circuit is the work done to move a unit charge from one point to the other.

Potential difference (V) between two points = Work done (W)/Charge (Q).

$$V = W/Q \quad \text{or} \quad W = VQ$$

- SI unit of potential difference is **volt (V)**, named after **Alessandro Volta** (Italy, 1745–1827).



ELECTRIC POTENTIAL AND POTENTIAL DIFFERENCE

One volt is the **potential difference** between two points in a current carrying conductor when **1 joule of work** is done to move **1 coulomb charge** from one point to other.

$$\text{Therefore, } 1 \text{ volt} = \frac{1 \text{ joule}}{1 \text{ coulomb}} \text{ or } 1 \text{ V} = 1 \text{ J C}^{-1}$$

Voltmeter: Instrument to measure potential difference. It is always connected in parallel across points between which the potential difference is to be measured.



Problem

How much work is done in moving a charge of 2 C across two points having a potential difference 12 V?

Solution

Amount of charge $Q = 2 \text{ C}$.

Potential difference $V = 12 \text{ V}$.

Amount of work $W = VQ$.

$$= 12 \text{ V} \times 2 \text{ C}$$

$$= \underline{24 \text{ J.}}$$

CIRCUIT DIAGRAM

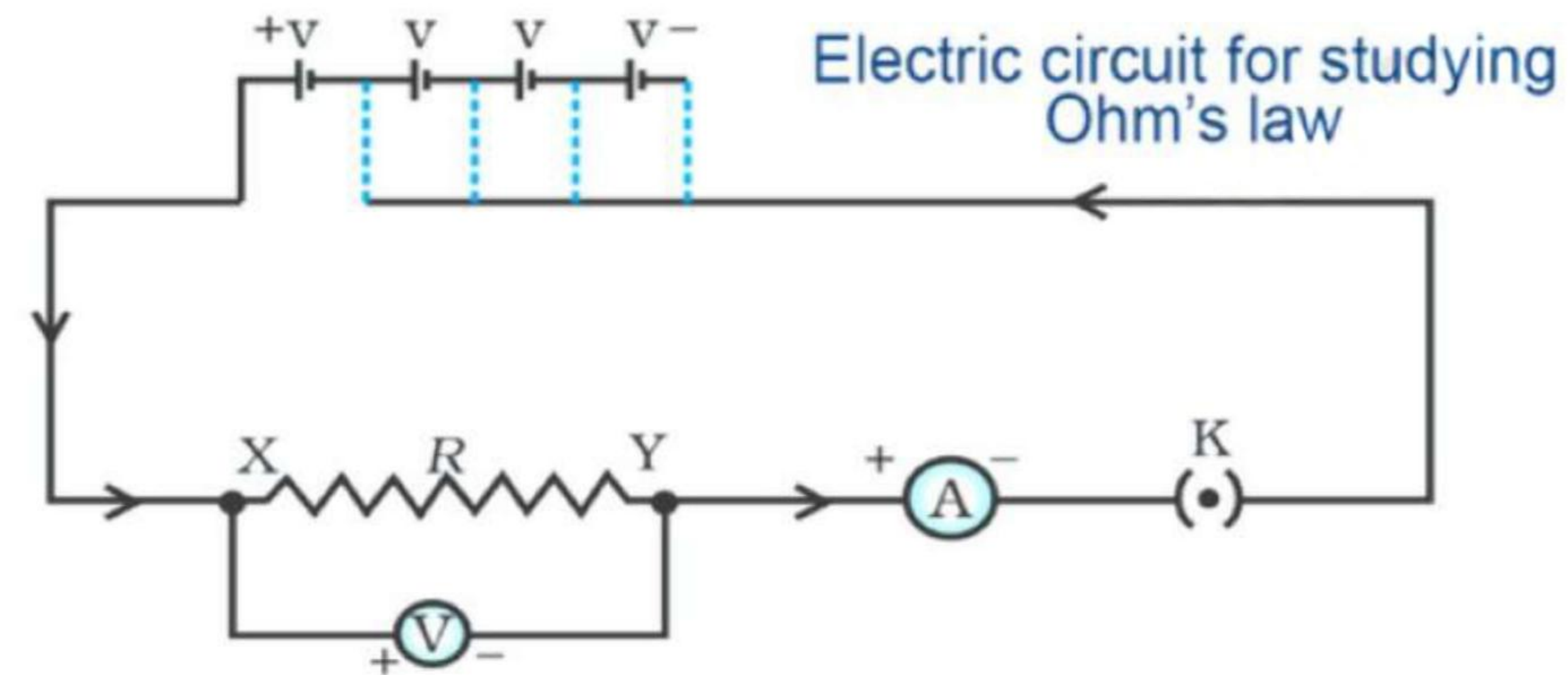
Conventional symbols of some commonly used components in circuit diagrams



Sl. No.	Components	Symbols
1	An electric cell	
2	A battery or a combination of cells	
3	Plug key or switch (open)	
4	Plug key or switch (closed)	
5	A wire joint	
6	Wires crossing without joining	
7	Electric bulb	
8	A resistor of resistance R	
9	Variable resistance or rheostat	
10	Ammeter	
11	Voltmeter	

OHM'S LAW

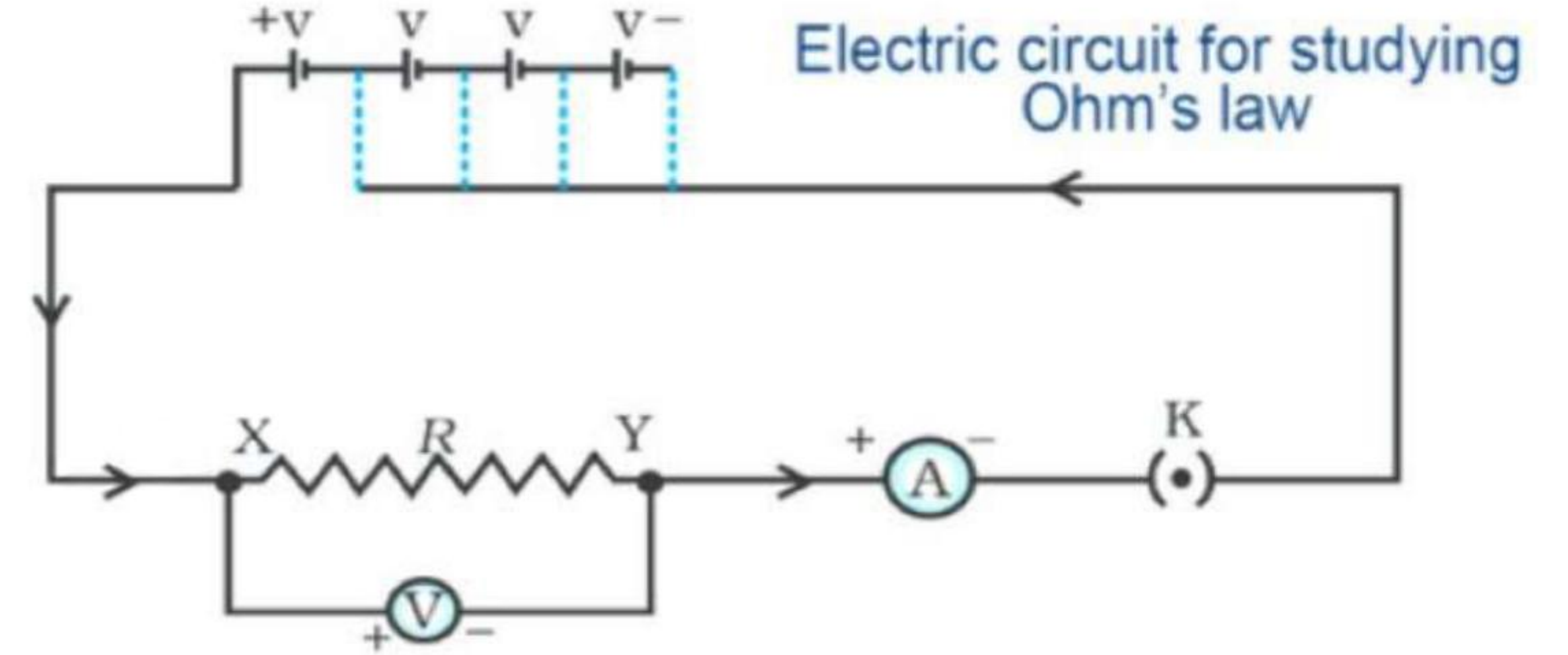
- Set up a circuit consisting of a nichrome wire XY (0.5 m length), an ammeter, a voltmeter and four cells of 1.5 V each. (Nichrome = an alloy of Ni, Cr, Mn & Fe.)
- Using one cell, note the ammeter reading I for the current and voltmeter reading V for potential difference across the nichrome wire. Repeat this using 2, 3 & 4 cells.



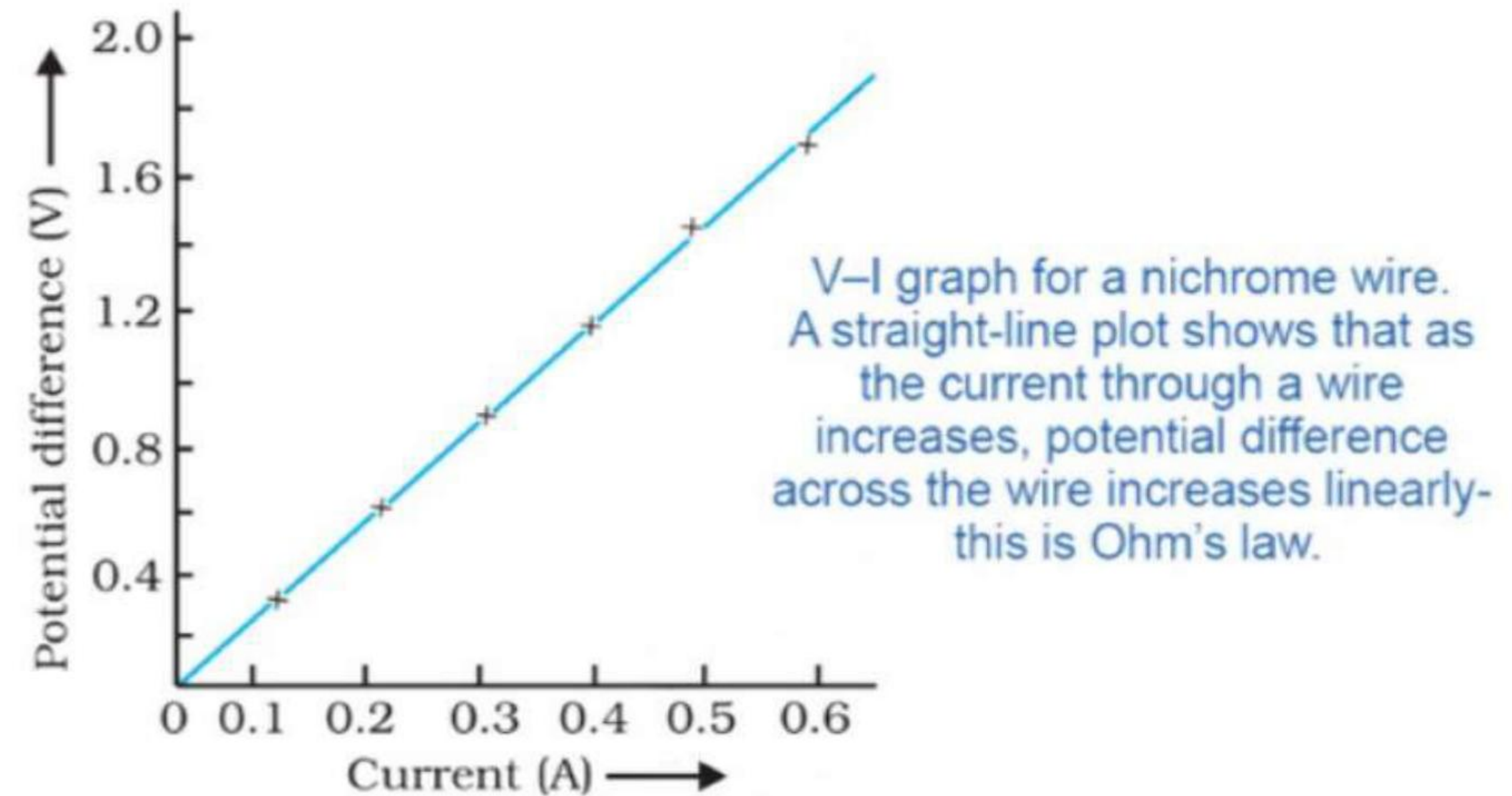
Number of cells used	Current through the nichrome wire, I (ampere)	Potential difference across nichrome wire, V (volt)	V/I (volt/ampere)
1	0.5	1.5	3
2	1.0	3.0	3
3	1.5	4.5	3
4	2.0	6.0	3

OHM'S LAW

- In each case, **V/I value** is approximately the same.
- **$V-I$ graph** is a straight line. Thus, V/I is a constant ratio.



Number of cells used	Current through the nichrome wire, I (ampere)	Potential difference across nichrome wire, V (volt)	V/I (volt/ampere)
1	0.5	1.5	3
2	1.0	3.0	3
3	1.5	4.5	3
4	2.0	6.0	3



OHM'S LAW

- In 1827, **Georg Simon Ohm** (Germany, 1787–1854) found out relationship between current and the potential difference.
- **Potential difference (V)** across the ends of a metallic wire in an electric circuit is directly proportional to the **current I** flowing through it at constant temperature. It is called **Ohm's law**.

$$V \propto I$$

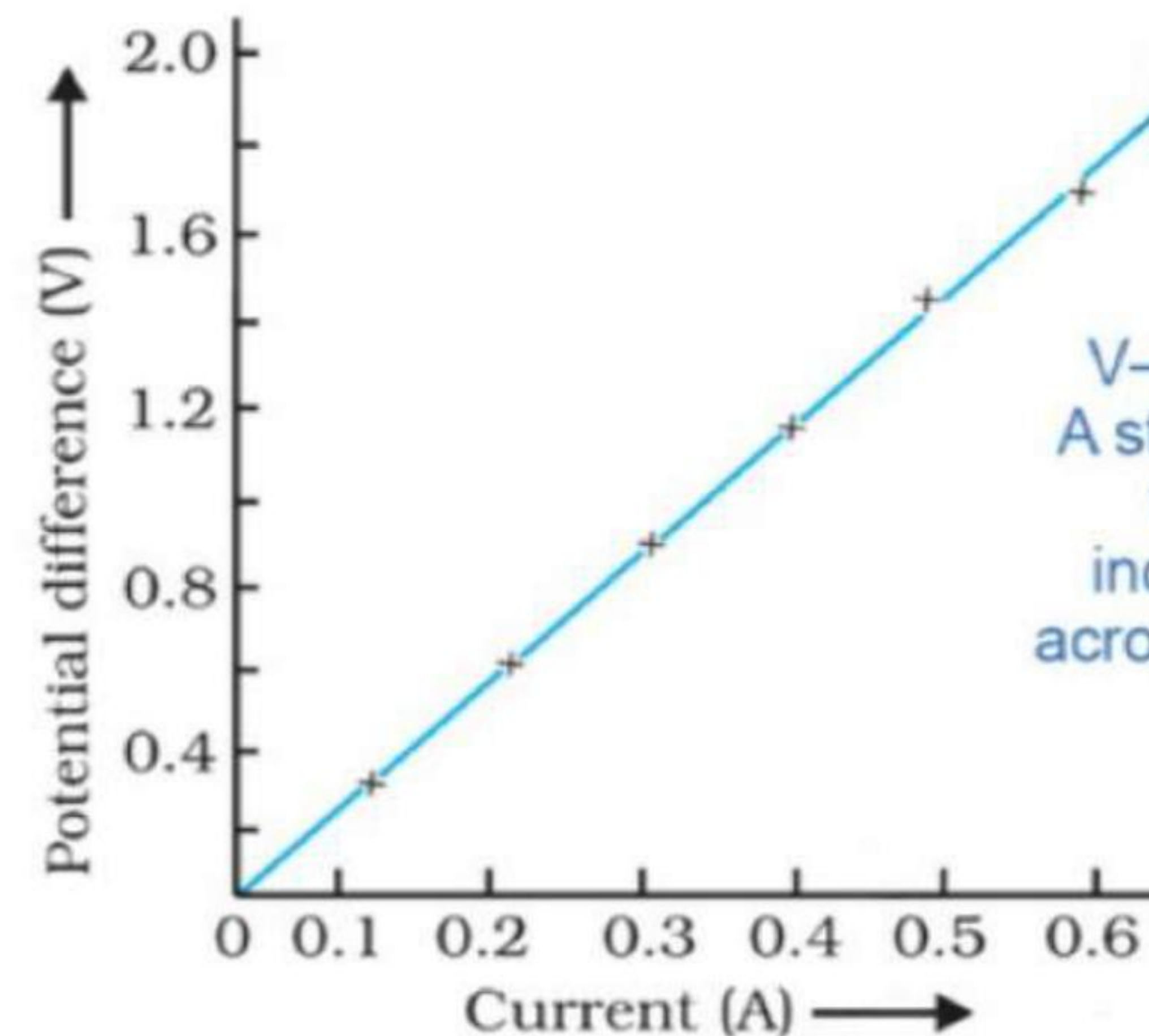
$$\text{or } V/I = \text{constant} = R$$

$$\text{or } V = IR$$

$$\text{or } I = V/R$$



Georg Simon Ohm



V-I graph for a nichrome wire. A straight-line plot shows that as the current through a wire increases, potential difference across the wire increases linearly- this is Ohm's law.

OHM'S LAW

- **R is a constant** for the given metallic wire at a given temperature and is called its **resistance**. It is the property of a conductor to resist the flow of charges. Its SI unit is **ohm (Ω)**.
- According to Ohm's law, $R = V/I$
- If the **potential difference** is **1 V** and the **current** is **1 A**, then the **resistance (R)** of the conductor is **1 Ω** .

$$1 \text{ ohm} = \frac{1 \text{ volt}}{1 \text{ ampere}}$$

- A conductor having appreciable resistance is called a **resistor**. It is used **to control electric current**.
- The current through a resistor is inversely proportional to its resistance. If the resistance is doubled the current gets halved. It has many practical applications.
- A component used to regulate current without changing the voltage source is called **variable resistance**. **Rheostat** is a variable resistor used to change resistance.



Resistors

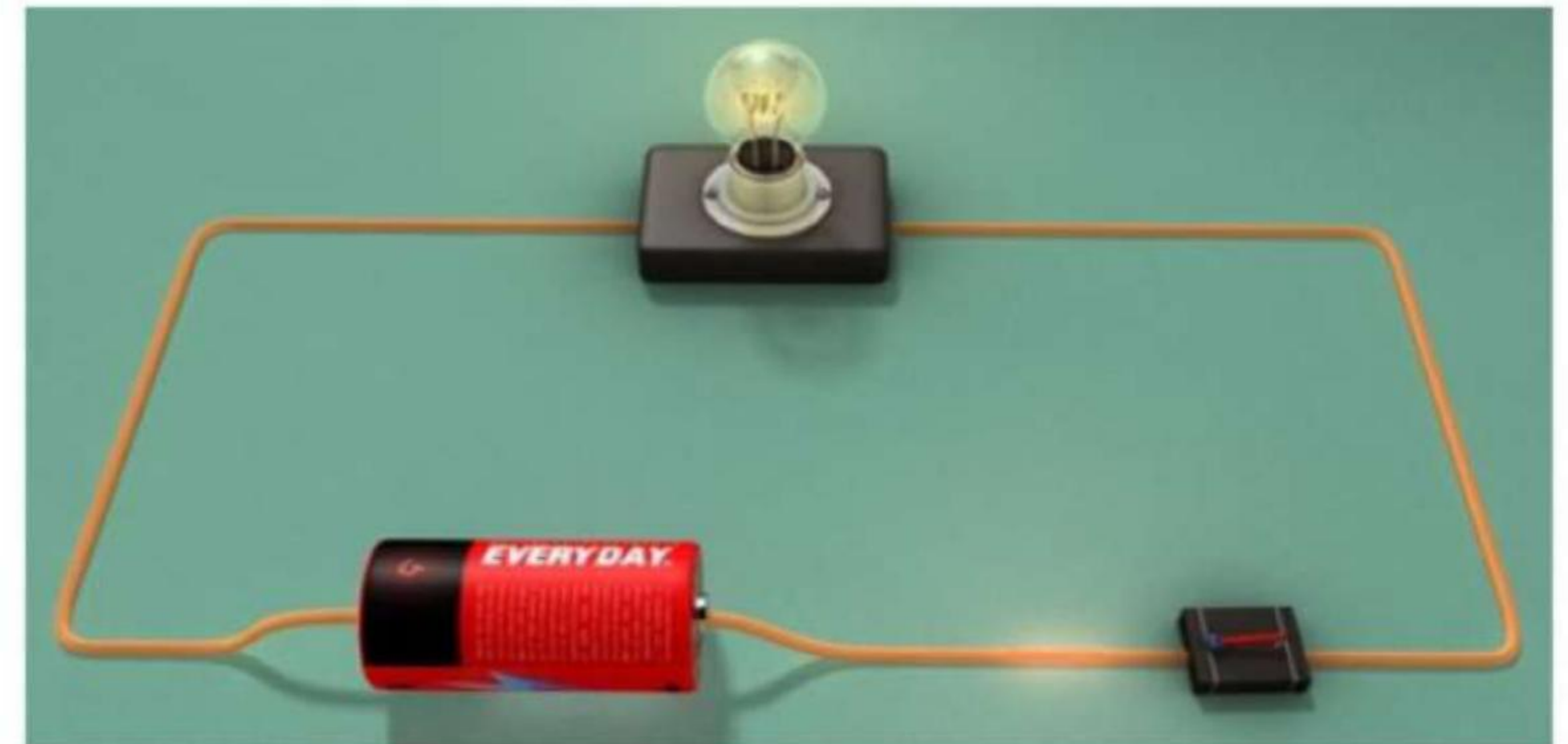
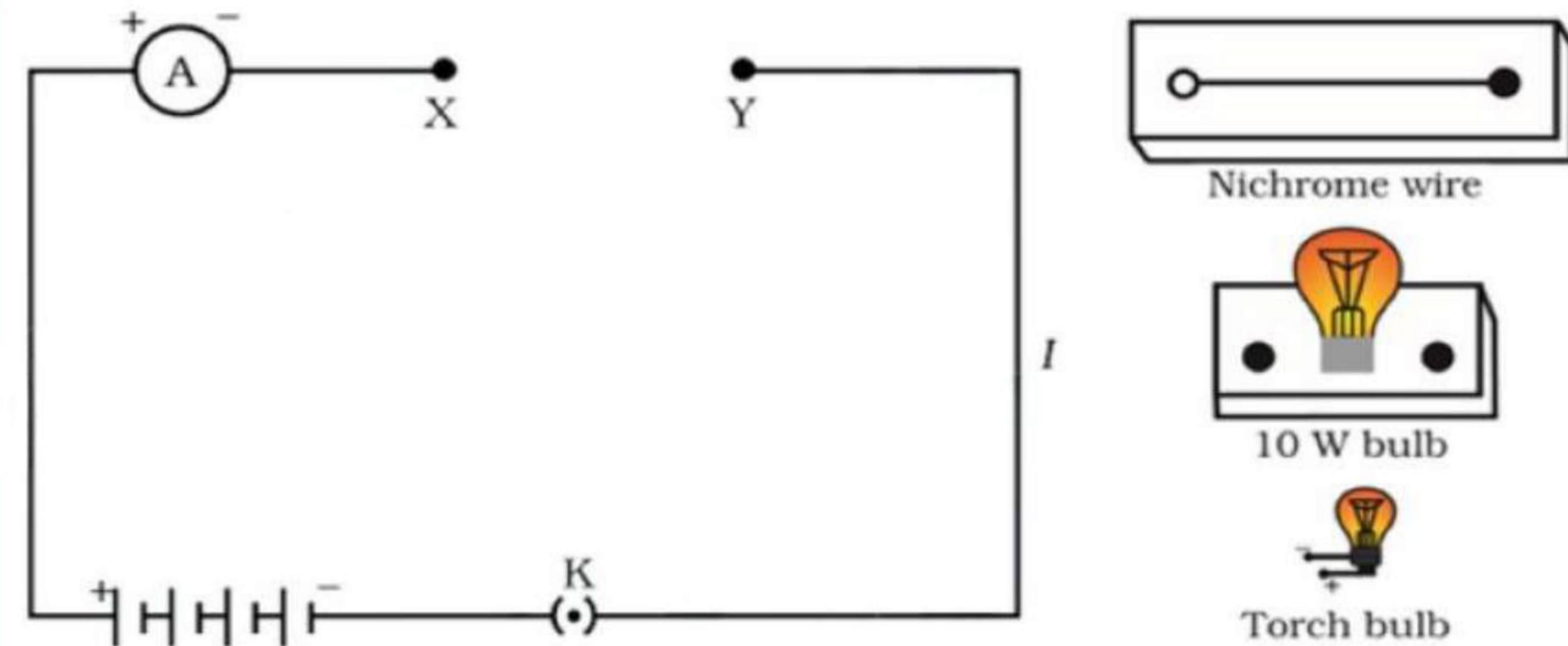


Rheostat

OHM'S LAW

Electrical resistance of a conductor

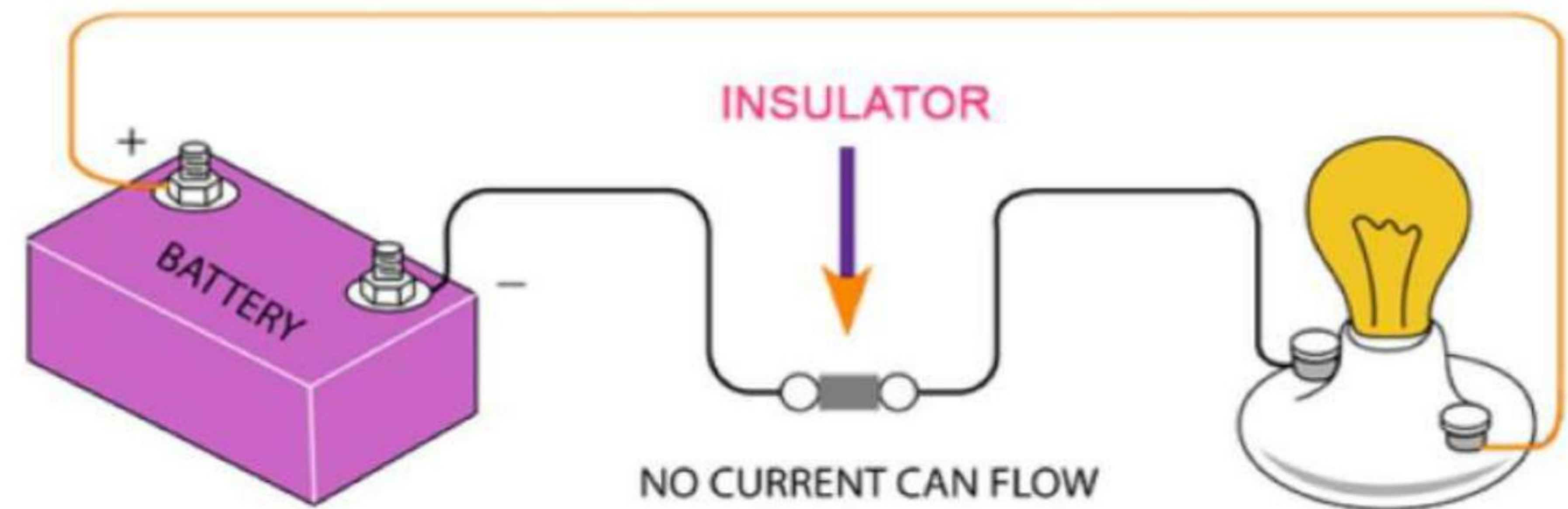
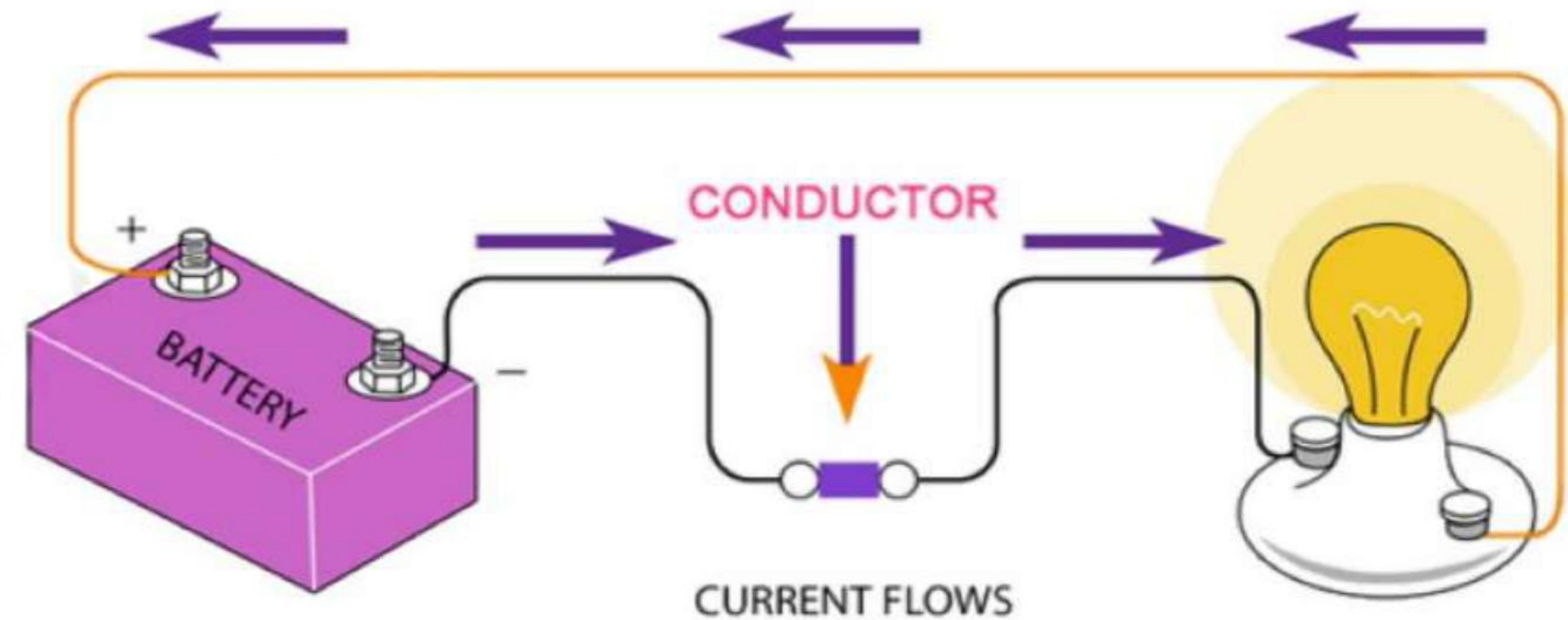
- Set up a circuit by connecting **four dry cells** of **1.5 V** each in series with the **ammeter** (0 – 5 A range) leaving a **gap XY** in the circuit.
- Complete the circuit by connecting a **nichrome wire** in the gap XY. Plug the key. Note down ammeter reading.
- Replace the nichrome wire with a **torch bulb** in the circuit and note down the ammeter reading.
- Repeat this with a **10 W bulb** and any material.
- The current is different for different components. In certain components, there is easy flow of electric current while the others resist the flow.



OHM'S LAW

Electrical resistance of a conductor

- Within a conductor, electrons are not completely free to move due to the attraction by atoms. So, resistance is increased and motion of electrons is retarded.
- A component that conducts electricity and has a low resistance is called a **good conductor**.
- A component with a high resistance is a **poor conductor**.
- A component with very high resistance is called an **insulator**. It does not conduct electricity.



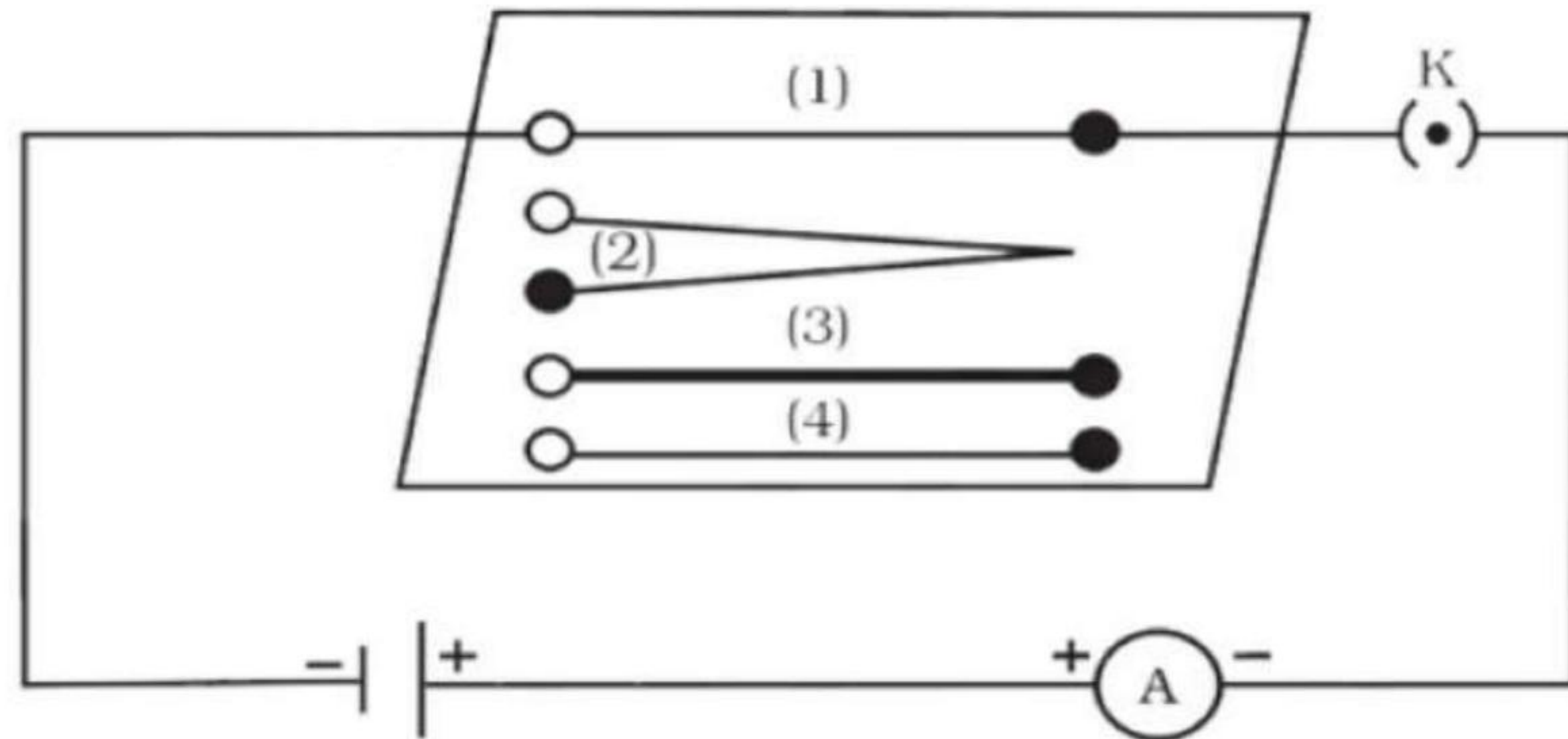
FACTORS ON WHICH THE RESISTANCE OF A CONDUCTOR DEPENDS

Note the **ammeter reading** in an electric circuit with

1. A **nichrome wire** of **length l** . Assume the reading = **1 A**.
2. **Nichrome wire** of twice the length (**$2l$**). Here, ammeter reading decreases to one-half (**0.5 A**).
3. A **thicker** (larger cross-sectional area) **nichrome wire** of the same **length l** . Here, reading is increased. If the area is doubled, reading is also doubled (**2A**).
4. A **copper wire** of same length and cross-sectional area as that of first nichrome wire. Here, reading is changed.



Nichrome wire



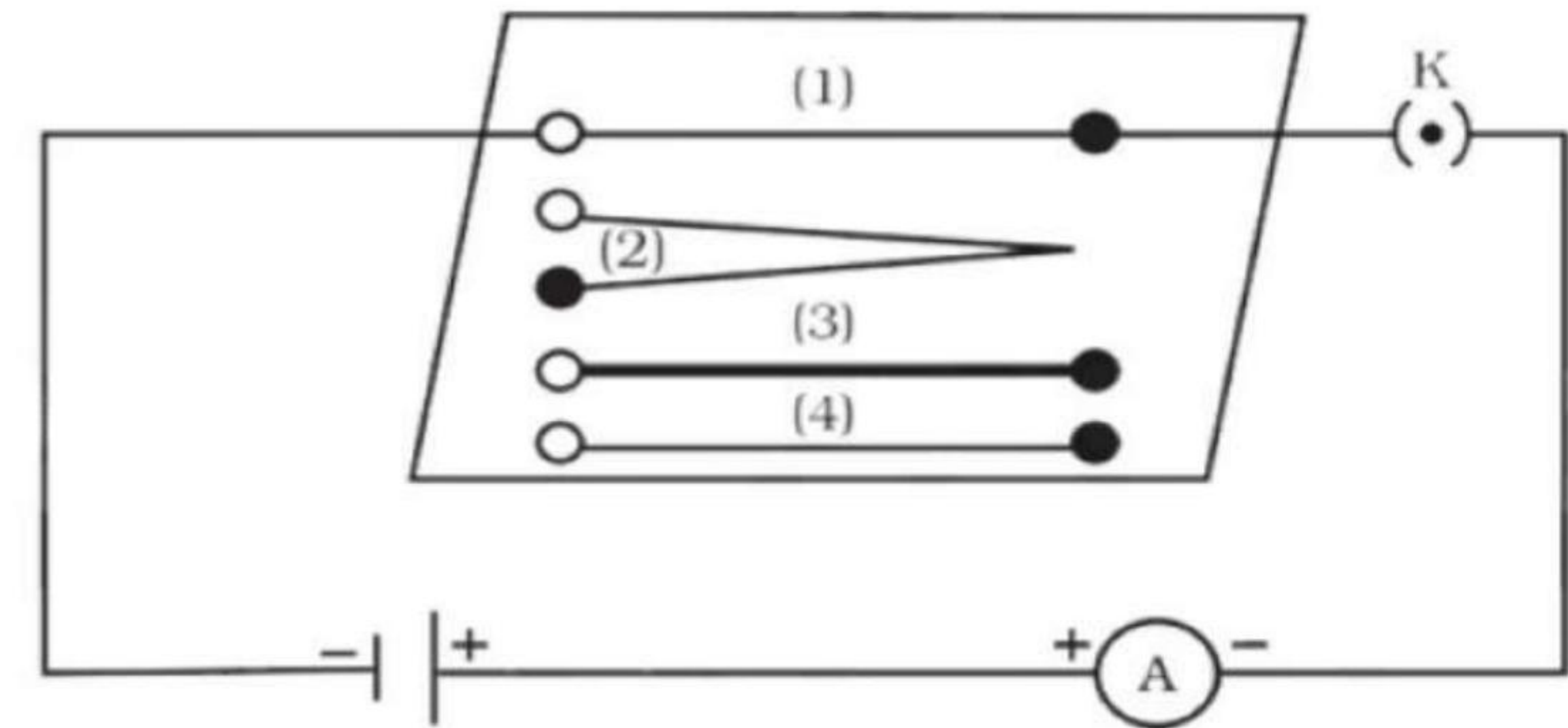
- _____ Nichrome wire, length l
- _____ Nichrome wire, length $2l$
- _____ Nichrome wire, thicker
- _____ Copper wire, length l

FACTORS ON WHICH THE RESISTANCE OF A CONDUCTOR DEPENDS

- Thus, resistance of a conductor depends on
 - i. Its length
 - ii. Its area of cross-section
 - iii. Nature of material.
- Resistance of a uniform metallic conductor is directly proportional to its length ($R \propto l$) and inversely proportional to the area of cross-section ($R \propto 1/A$). i.e.,

$$R \propto \frac{l}{A} \quad \text{or} \quad R = \rho \frac{l}{A}$$

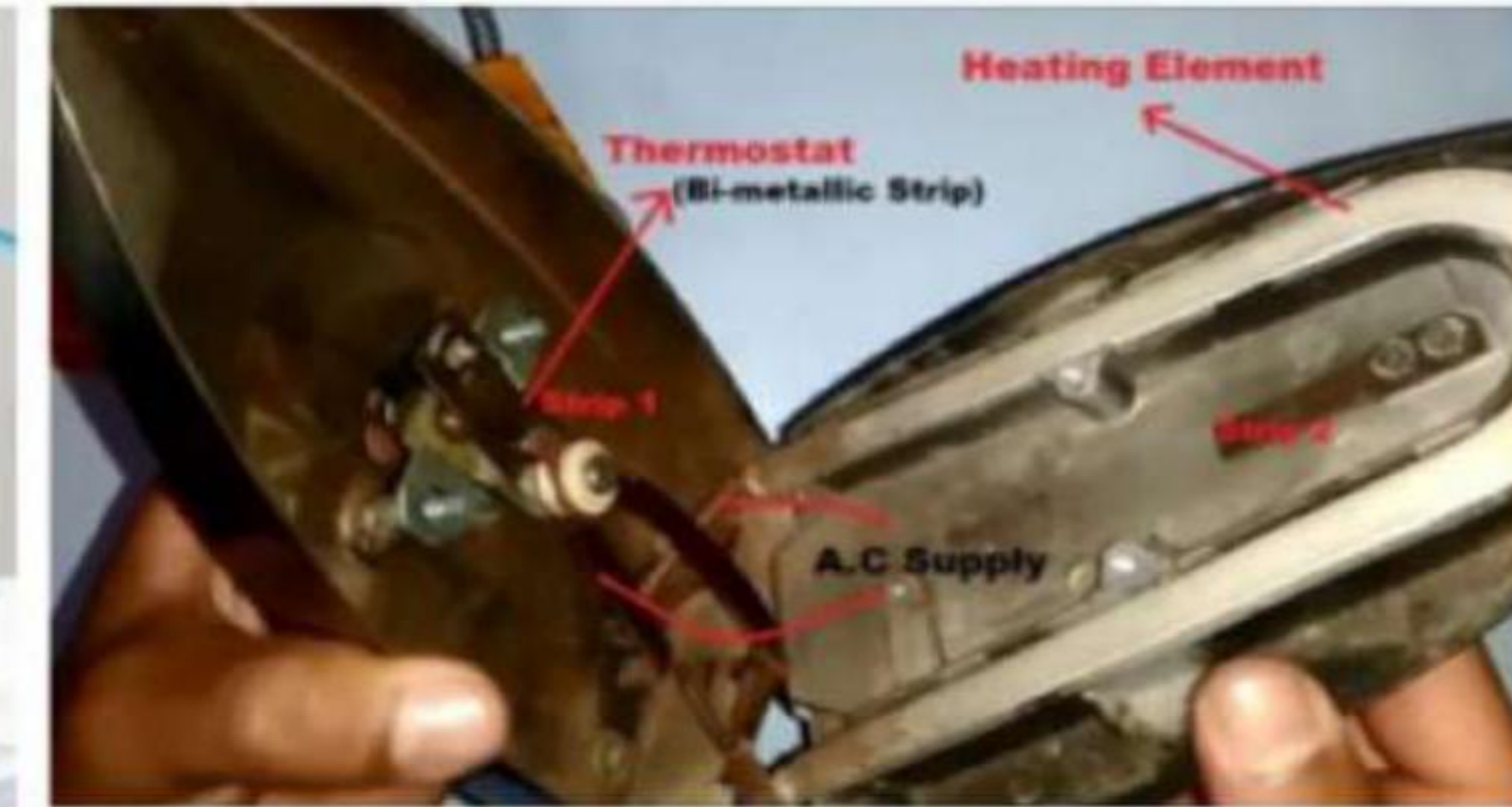
- Where ρ (rho) is a **constant** of proportionality and is called **electrical resistivity** of the material of conductor. It is a characteristic property of the material.
- SI unit of resistivity is $\Omega \text{ m}$.



- _____ Nichrome wire, length l
- _____ Nichrome wire, length $2l$
- _____ Nichrome wire, thicker
- _____ Copper wire, length l

FACTORS ON WHICH THE RESISTANCE OF A CONDUCTOR DEPENDS

- Resistivity of metals & alloys is very low (10^{-8} to $10^{-6} \Omega \text{ m}$). They are **good conductors** of electricity.
- Resistivity of insulators (rubber, glass etc.) is **very high** (10^{12} to $10^{17} \Omega \text{ m}$).
- Resistance and resistivity vary with temperature.
- Resistivity of an **alloy** is generally higher than that of its constituent metals. Alloys do not oxidise (burn) readily at high temperatures. So, they are used in **electrical heating devices** like electric iron, toasters etc.
- **Tungsten** is used for filaments of electric bulbs. **Copper & aluminium** are used for electrical transmission lines.



FACTORS ON WHICH THE RESISTANCE OF A CONDUCTOR DEPENDS

Electrical
resistivity of
some substances
at 20°C

	Material	Resistivity ($\Omega \text{ m}$)
Conductors	Silver	1.60×10^{-8}
	Copper	1.62×10^{-8}
	Aluminium	2.63×10^{-8}
	Tungsten	5.20×10^{-8}
	Nickel	6.84×10^{-8}
	Iron	10.0×10^{-8}
	Chromium	12.9×10^{-8}
	Mercury	94.0×10^{-8}
	Manganese	1.84×10^{-6}
Alloys	Constantan (Cu + Ni)	49×10^{-6}
	Manganin (Cu + Mn + Ni)	44×10^{-6}
	Nichrome (Ni + Cr + Mn + Fe)	100×10^{-6}
Insulators	Glass	$10^{10} - 10^{14}$
	Hard rubber	$10^{13} - 10^{16}$
	Ebonite	$10^{15} - 10^{17}$
	Diamond	$10^{12} - 10^{13}$
	Paper (dry)	10^{12}

FACTORS ON WHICH THE RESISTANCE OF A CONDUCTOR DEPENDS

Problem

- a) How much current will an electric bulb draw from a 220 V source, if the resistance of bulb filament is $1200\ \Omega$?
- b) How much current will an electric heater draw from a 220 V source, if the resistance of heater coil is $100\ \Omega$?

Solution

a) $V = 220\text{ V}; R = 1200\ \Omega.$

Current, $I = V/R = 220\text{ V}/1200\ \Omega = \underline{\underline{0.18\text{ A}}}.$

b) $V = 220\text{ V}, R = 100\ \Omega.$

Current, $I = 220\text{ V}/100\ \Omega = \underline{\underline{2.2\text{ A}}}.$

Thus the current drawn by an electric bulb and electric heater from same 220 V source is different.

Problem

Potential difference between the terminals of an electric heater is 60 V when it draws a current of 4 A from the source. What current will the heater draw if the potential difference is increased to 120 V?

Solution

Potential difference $V = 60\text{ V}$, current $I = 4\text{ A}.$

$$R = V/I = (60\text{ V})/(4\text{ A}) = \underline{\underline{15\ \Omega}}.$$

When the potential difference is increased to 120 V

Current $I = V/R = (120\text{ V})/(15\ \Omega) = \underline{\underline{8\text{ A}}}.$

FACTORS ON WHICH THE RESISTANCE OF A CONDUCTOR DEPENDS

Problem

Resistance of a metal wire of length 1 m is $26\ \Omega$ at 20°C . If the diameter of the wire is 0.3 mm, what will be the resistivity of the metal at that temperature? Predict the material of the wire.

Solution

- Resistance R of the wire = $26\ \Omega$
- Diameter $d = 0.3\ \text{mm} = 3 \times 10^{-4}\ \text{m}$
- Length l of the wire = 1 m.
- Resistivity of the metallic wire,
 $\rho = (RA/l) = (R\pi d^2/4l) = \underline{1.84 \times 10^{-6}\ \Omega\ \text{m}}$
- This is the resistivity of manganese.

Problem

A wire of given material having length l and area of cross-section A has a resistance of $4\ \Omega$. What would be the resistance of another wire of the same material having length $l/2$ and area of cross-section $2A$?

Solution

For first wire: $R_1 = \rho l/A = 4\ \Omega$

For second wire: $R_2 = \rho \frac{l/2}{2A} = \frac{1}{4} \rho l/A$

$$R_2 = \frac{1}{4} R_1 = \underline{1\ \Omega}$$